AI Your Way

Compute-Agnostic, Rack-Scale Architecture for AI Factories

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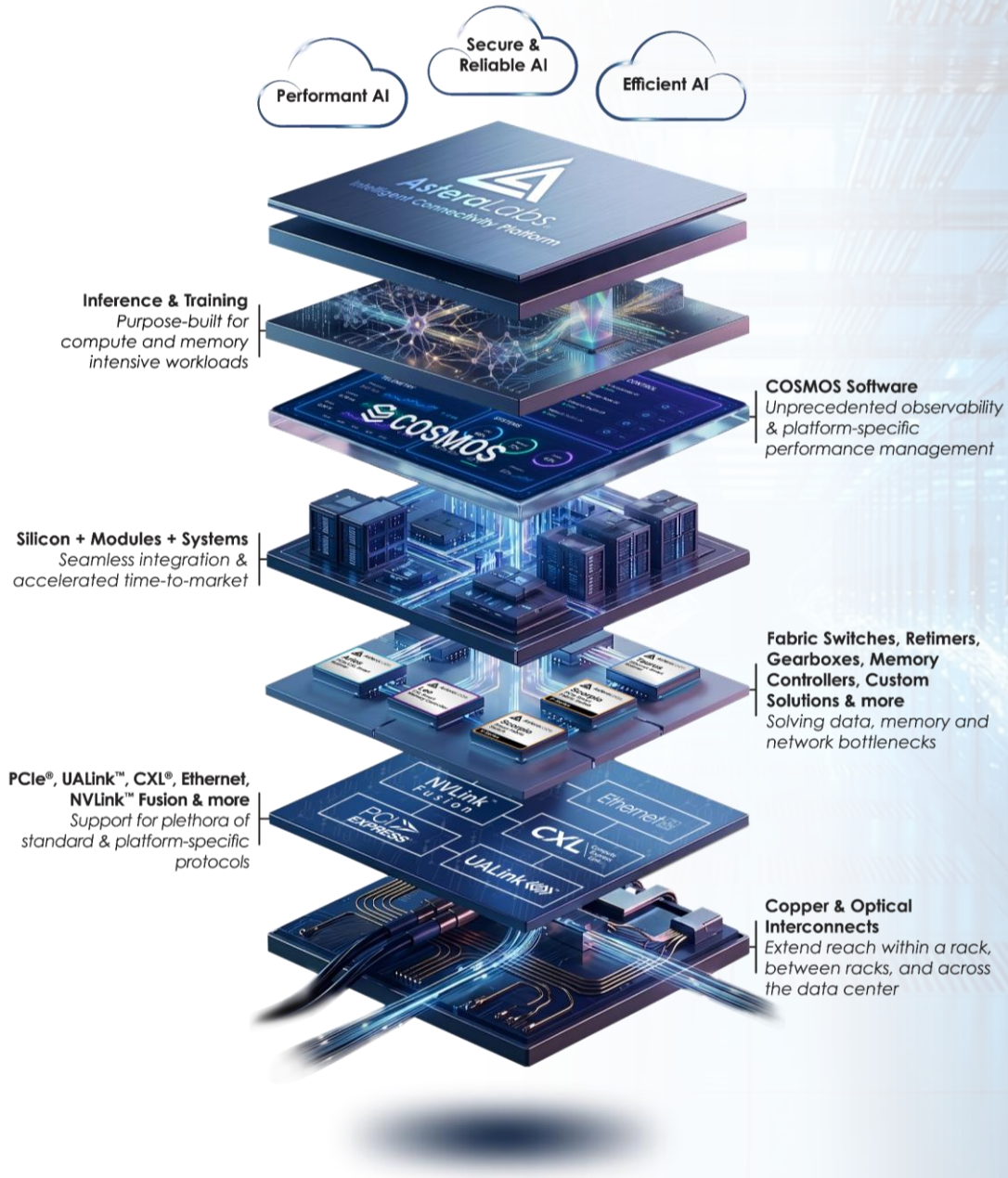
October 12, 2026

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AI Your Way.

Purpose-built rack-scale connectivity.

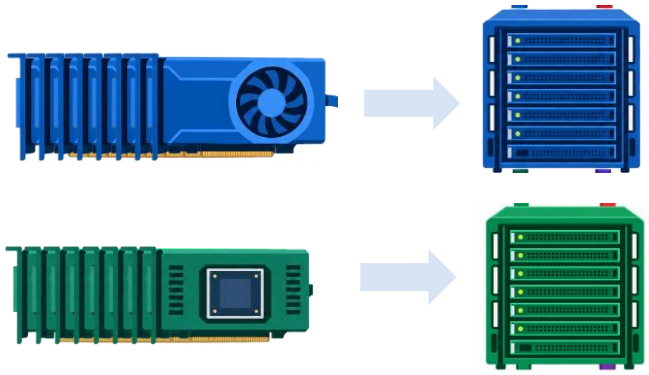
any Workload Accelerator



Industry Challenges for Rack-Scale Deployment

Challenge 1

Unprecedented Demand for Compute

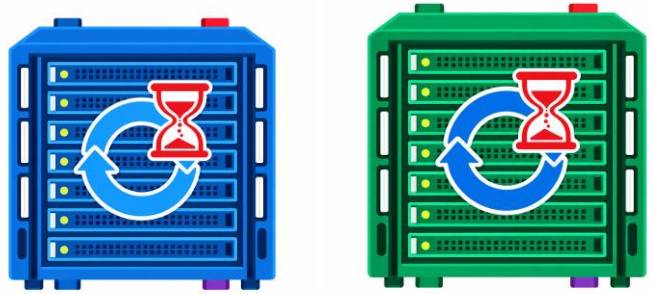


Significant supply constraints

- Workloads increasingly favor specialized xPUs
- Availability, lead time & cost shifts
- Platform-specific trays & cabling

Challenge 2

Rack-Scale Deployment & Management

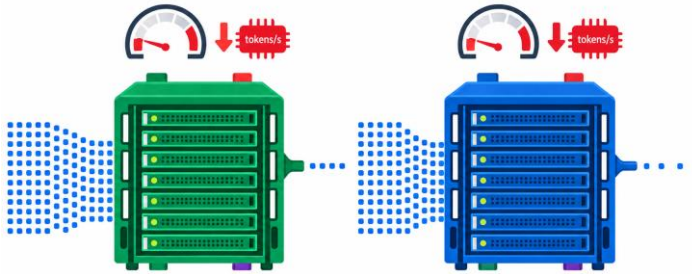


New xPU ➔ New Rack!!

- Re-integration for each new xPU
- Repeated validation & testing at rack-scale
- Capital prohibitive – Enterprise AI, Inference

Challenge 3

Low-Latency Inference Drives Agentic AI



Lower tokens/s ➔ higher cost per token

- Connectivity bottlenecks cap tokens/s per xPU
- Low latency needed at scale
- KV-cache growth strains memory

Industry trends demand rack-scale architectures that deploy faster & maximize tokens

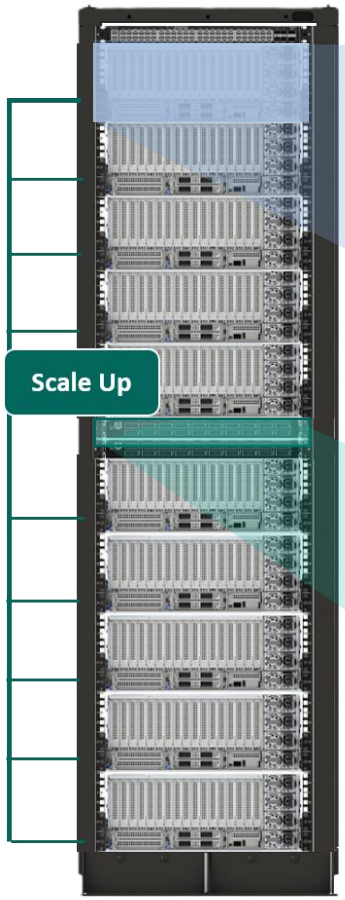


Introducing Orion

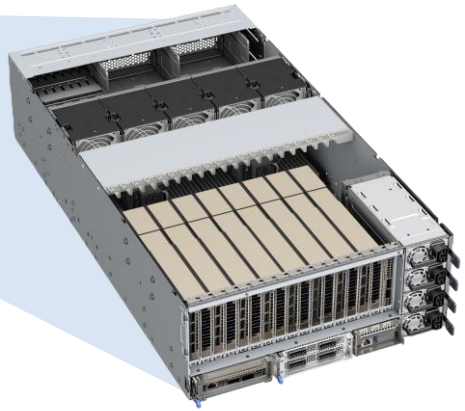
Compute-Agnostic Rack-Scale Reference Architecture

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Orion: Production-Grade Rack-Scale Reference Design



Orion *Rack*
Scale-Up GPUs + Memory



Orion-CT
xPU-agnostic ***Compute Tray***



Orion-FT
Scale-Up ***Fabric Tray***

Compute-Agnostic Architecture

1

- Select best-of-breed accelerators based on workloads
- Common Orion-CT, **compute tray**, for accelerators & memory
- Leverage Orion-FT**, fabric tray**, as a common scale-up fabric

Production-Grade: Rack-Scale Reference Architecture

2

- Co-developed with ODM partners
- Available for resale via OEM partners
- Enhanced **COSMOS** to include fabric & rack management

Fabric-Efficient Token Economics

3

- Enabled by Astera Labs' **Scorpio X-Series, Leo X-Series, COSMOS**
- Leverage Orion-MT, **memory tray,** for KV Cache acceleration
- In-Network-Compute (INC) & Hypercast™ accelerates collectives

AI Your Way: Orion turns any GPU into a deployment-ready and token-efficient solution

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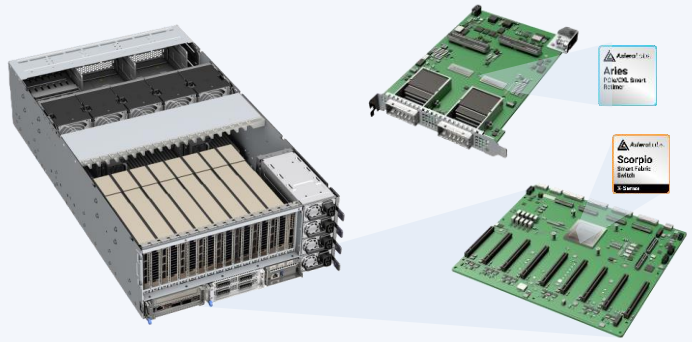
Orion: PCIe 6 Modular Reference Design for Diverse Workloads

8 GPUs → **160 GPUs**

Orion-CT

*xPU-Agnostic **Compute Tray**, 4RU*

- Enabled by **Scorpio 320L, X-Series**
- 8 GPUs scale-up with **Orion-CB**
- Reach extension for 8+ GPUs with **Orion-LB**



Orion-FT

*PCIe Scale-Up **Fabric Tray**, 1RU*

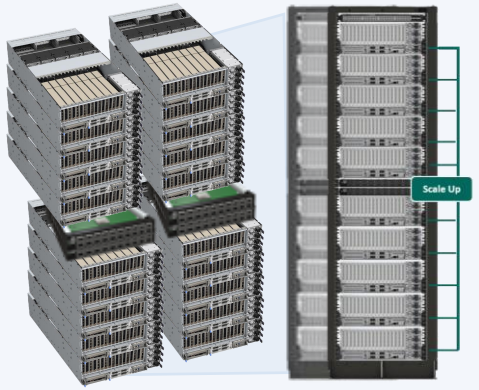
- Enabled by **Scorpio 320L, X-Series**
- Aggregates all GPUs in **Orion-CT**
- Non-blocking bandwidth across fabric



Orion Rack

160 GPUs / POD, 2 Racks

- Enhanced **COSMOS:** Fabric & Rack Mgr
- 160 GPUs scale-up, non-blocking
- Cabling solutions: copper, optics



COSMOS

Modular compute-agnostic architecture to scale up from 8 to 160 accelerators


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It Starts with the RIGHT Scale-Up Fabric…

Requirements

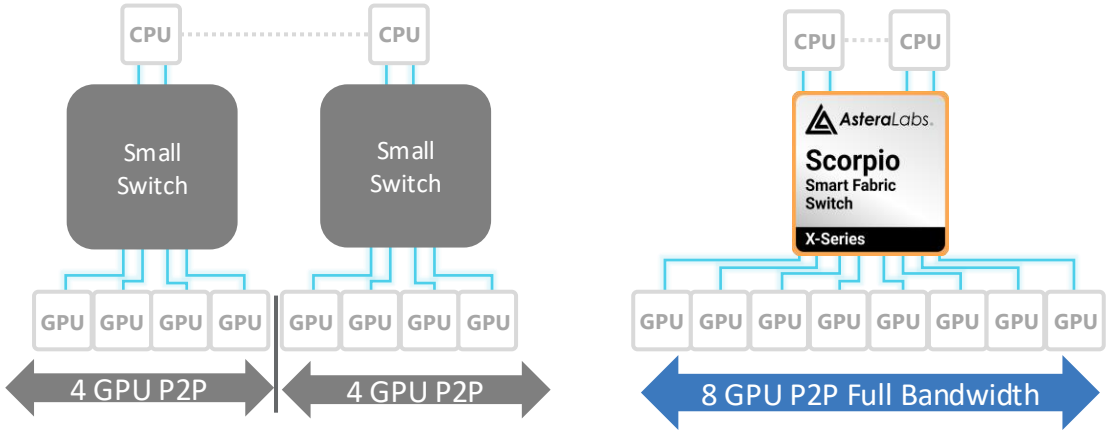
- Universal, Memory Semantic → PCIe-based, Platform-specific
- Scale, Bandwidth, Low Latency → Largest Radix
- Maximize GPU Utilization → Hypercast™ & In-Network Compute
- Always-on & Resilient → Multi-plane, advanced fabric RAS



Scorpio X-series 320L

Industry's largest PCIe switch now in volume production shipping to multiple AI infrastructure providers

Size and Intelligence Matter



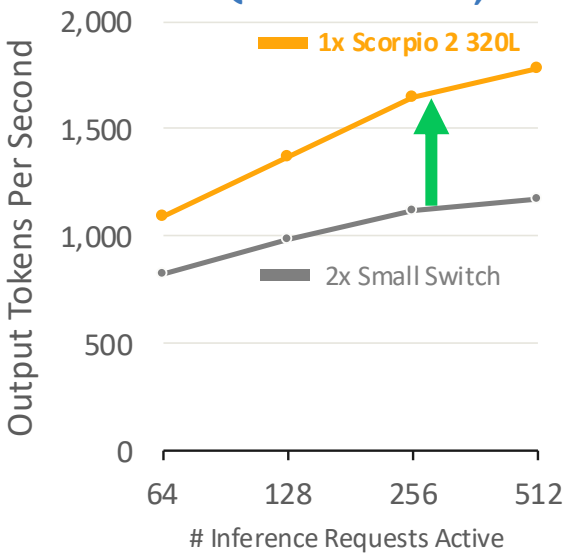


Hypercast + **INC**

Data Replication Engine
Accelerates one-to-many

In-Network Compute
Accelerates many-to-one

Qwen3.5-122B, Balanced Inference Prefill/Decode



+50% Tokens Per Second
+90% in Prefill heavy workloads

- ↑ GPU utilization
- $ Millions rack-scale value



Intelligence accelerate collective operations by up to 2x
Faster routing and collectives boost response and Tokens per Second

- ↑ GPU utilization
- $ Millions rack-scale value



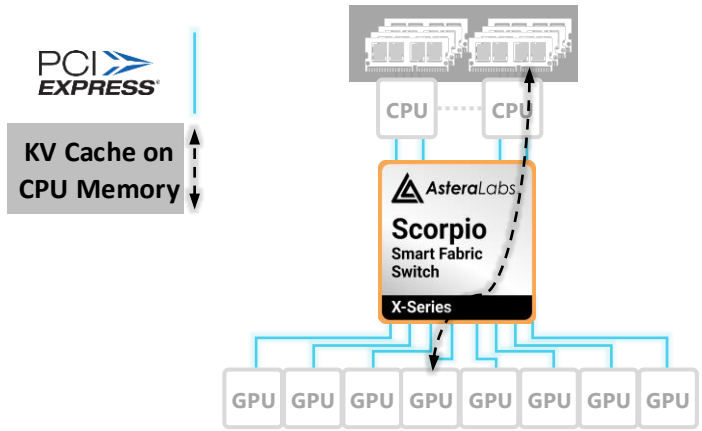
Do More with Less: The ultimate differentiator in a constrained environment

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Leo X-Series Maximizes Token Economics

Where the Bottleneck Sits for Inference Workloads

Prefill is compute-bound. **Decode** is memory-bandwidth-bound and long context with multi-turn, concurrent users make it the dominant phase.

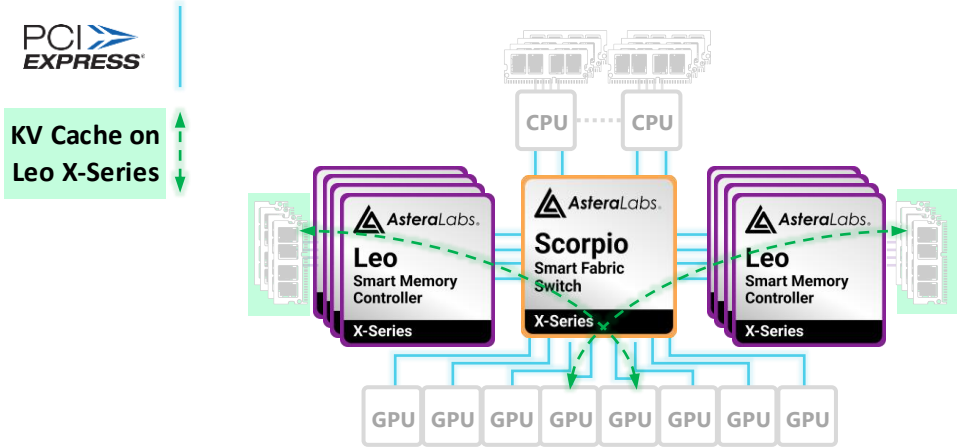


- KV Cache quickly exceeds local GPU memory capacity
- KV Cache is offloaded to CPU DRAM, NVMe storage or the network, but **adds latency and bandwidth bottlenecks**

Leo X-Series Performance*

62% Faster
time to first token (TTFT)

22% More
tokens per second (TPS)



- **Leo X-Series + Scorpio Smart Fabric Switches provide a dedicated KV Cache memory tier near the GPU**
- Supports any PCIe and other platform-specific based GPUs
- Scale-up KVcache capacity independent of CPUs & GPUs

Leo X-Series and Scorpio X-Series form a tightly coupled compute & memory scale-up solution for Agentic AI

** As measured on an Intel GNR server with four NVIDIA H200 GPUs on Qwen2.5-32B AI Model*

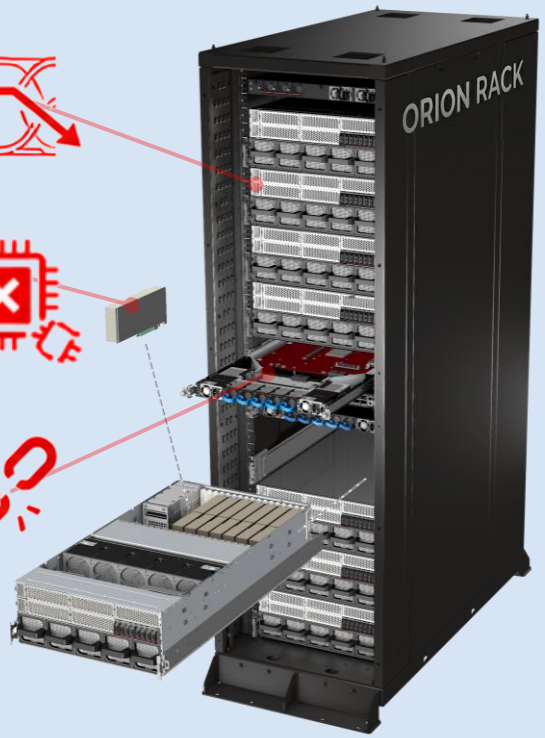
COSMOS Enhancements – Device Level to Rack Level

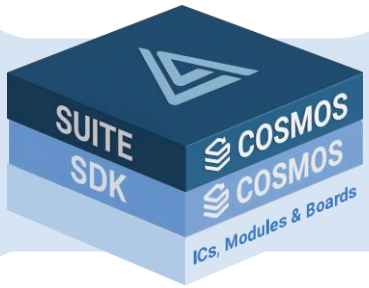
Challenges of Rack and Fabric Management

Signal Degradations
Rising FEC/BER, link recoveries, replays, degrading margins

GPU Link Drops
Isolates link segments, partners, stalling workloads

Fabric Link Failures
Switch-to-switch PCIe link failure puts the whole fabric at risk





COSMOS Rack & Fabric Manager

Rack Capabilities

- Unified rack-scale visibility across compute, link, and switches
- Standardized APIs for monitoring, management, and automation
- System health, inventory, and rack-level orchestration

Fabric Capabilities

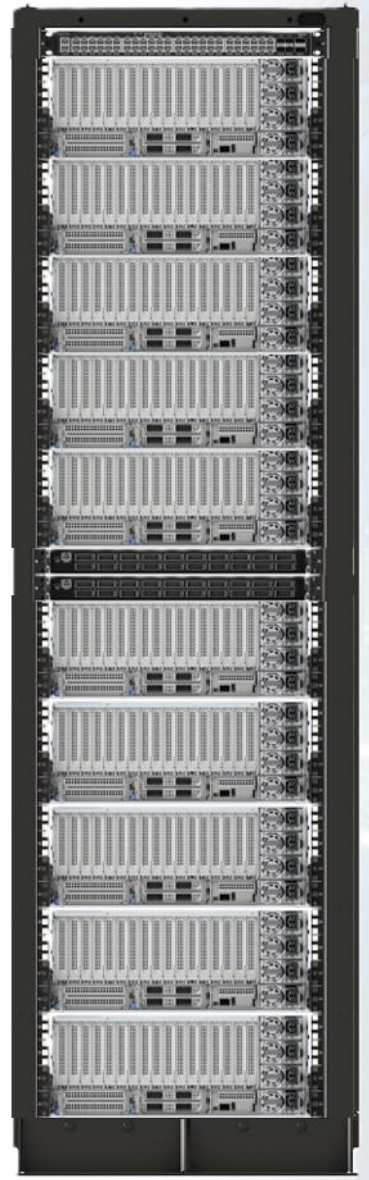
- End-to-end fabric topology, offload engines, device mapping
- GPU, link, and switch health and performance monitoring
- Fault isolation, alerts, and automated remediation

Scorpio	Leo	Aries	Taurus
Smart Fabric Switch	Smart Memory Controller	PCIe/CXL Smart Retimer	Ethernet Smart Retimer

Unified telemetry for XPU connectivity, signal integrity, and fabric health

AI Your Way
with Orion Rack-Scale Reference Architecture

- Compute-agnostic
- Production-grade solution
- Modular scalable architecture
- Fabric-efficient token economics
- Enhanced COSMOS



See Orion in Astera Labs Booth B34 and Hyve Solutions Booth C31

Thank You

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Check us out on

www.asteralabs.com